

A Multi-Source Fusion Approach for Driver Fatigue Detection Using Physiological Signals and Facial Image

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Abstract—Detecting driver fatigue is critical to ensuring road safety. Existing fatigue detection methods typically rely on traditional hand-picked features as inputs. However, these hand-picked features can hardly respond accurately to the driver's fatigue state due to a certain degree of subjectivity and the extraction of these features requires a long time window, which limits the accuracy and real-time performance of the detection. This paper proposes a novel fatigue detection method based on multi-source information fusion, which relies entirely on neural networks for automatic feature extraction. Through simulated driving experiments, we recorded physiological signals and facial videos from 21 participants for model training and testing. The results show that our model outperforms existing methods in terms of accuracy and real-time performance, achieving a detection accuracy of 93.15% within a 3-second time window (specificity = 94.04%, sensitivity = 91.71%). The visualization results of the model reveal potential relationships between facial regions for the first time, validating the rationality and effectiveness of our method. The practical issues of fatigue detection methods and future research directions are also explored.

Index Terms—Fatigue detection, physiological signal, behavioral feature, convolutional neural network, multi-source information fusion.

I. INTRODUCTION

FATIGUE driving is one of the major causes of traffic accidents [1] with approximately 20% of accidents worldwide being attributed to driver fatigue [2]. These accidents result in hundreds of thousands of casualties annually [3]. While automated driving systems can mitigate the impact of human

factors on road safety to some extent, the presence of potential safety risks that automated systems may not effectively handle necessitates the readiness of drivers to take over control at any given moment [4], even in vehicles equipped with state-of-the-art automated driving systems. Moreover, research indicates that prolonged use of automated driving functions is more prone to inducing driver fatigue compared to manual driving [5], [6]. Therefore, real-time fatigue detection of drivers holds crucial significance for ensuring the safety of both conventional and conditionally automated vehicles.

In this context, researchers have proposed various real-time fatigue detection methods. Some studies have focused on detecting driver fatigue using physiological signals, such as EEG [7] and ECG [8]. However, physiological signal acquisition often requires devices to be closely attached to the driver's body, which can be highly invasive and directly impact the driver's comfort and normal vehicle control. Consequently, although fatigue detection based on physiological signals can achieve high accuracy, the practical application of such methods becomes challenging. Nevertheless, with the rapid development of wearable devices, smartwatches, and earphones introduced by major companies can now capture physiological signals like heart rate and blood oxygen saturation. Moreover, these devices can achieve real-time data transmission with other electronic devices, showing strong potential for application in driver fatigue detection. Therefore, it is essential to focus on how to leverage these easily collectible physiological signals to detect the driver's fatigue state.

Some studies focus on the behavioral characteristics of drivers, in particular facial fatigue characteristics [9], [10]. Many hand-picked fatigue features have been proposed, such as the percentage of eyelid closure (PERCLOS), blink frequency, and eye openness level [11], [12], [13]. These features are mostly related to the eye and mouth regions [14], as the changes in these areas have a clear relationship with driver fatigue, making them convenient for the design of handcrafted features. However, relying solely on features related to the eye and mouth regions can easily overlook other information related to driver fatigue, including changes in facial expression and muscle movement, which is detrimental to improving the accuracy and robustness of fatigue detection. In addition, when drivers wear sunglasses or masks, these critical regions

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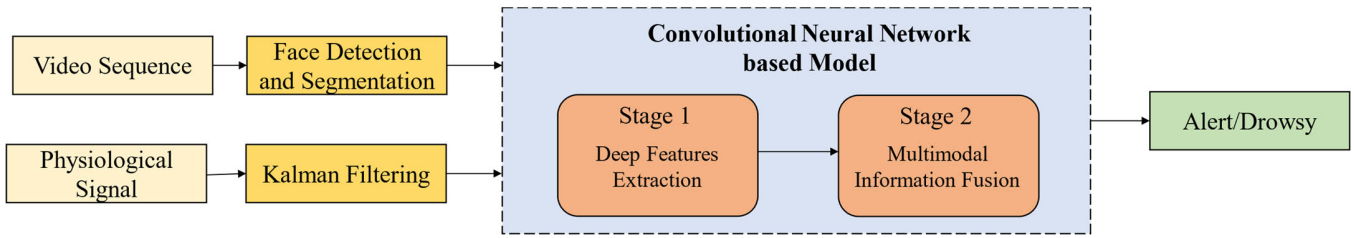


Fig. 1. Framework of the driver fatigue detection model.

can be obscured, rendering fatigue detection methods ineffective. Therefore, it is necessary to investigate the relationship between other facial regions and fatigue expression, and to use other facial regions for fatigue detection.

To overcome the limitations of fatigue detection methods based on single data sources in terms of detection accuracy and robustness, researchers typically perform fatigue detection by fusing multimodal features [15], [16]. With the rise of deep learning techniques, deep learning has become the primary method for constructing fatigue detection models based on information fusion [17], [18]. However, existing studies often use hand-picked features as inputs to deep learning models [19], [20] and fail to take advantage of deep learning models in automatic feature extraction. There are some drawbacks associated with using these hand-picked fatigue features. Firstly, these hand-picked features are designed by experts based on prior knowledge and are subjective, making it difficult to capture potential fatigue features. Secondly, the extraction of these hand-picked features typically requires a long time interval [21], [22], [23]. Therefore, the reliance on hand-picked features affects the effectiveness of fatigue detection and leads to several problems in terms of accuracy and real-time performance.

To address these issues, we propose an innovative fatigue detection method, as shown in Fig. 1. It takes the driver's entire facial image and physiological signals as input, and relies on neural networks for automatic feature extraction. The selected physiological signals can be collected using non-invasive devices, which facilitates the practical application of our method. In addition, we performed visual analysis on the model to validate the effectiveness of the extracted features. The main contributions of this paper are as follows:

1) A novel approach to fatigue detection is proposed that does not rely on traditional hand-picked features, but instead integrates physiological signals with the entire facial image for multi-source information fusion. It achieves a detection accuracy of 93.15% (specificity = 94.04%, sensitivity = 91.71%) within a 3-second time window, outperforming existing methods in both detection accuracy and real-time performance.

2) We prove the rationale and effectiveness of direct fatigue feature extraction using neural networks through visual analysis of the model. The model focused on key regions such as the eyes and mouth, as expected, and also identified additional potential fatigue features that are difficult to capture through handcrafted feature design.

3) For the first time, we reveal potential relationships between drivers' facial regions and fatigue. The results indicate

that in addition to the commonly studied areas such as the eyes and mouth, other facial regions such as the eyebrows and nose can also reflect the state of fatigue.

The remaining sections of this paper are organized as follows: Section II discusses previous related research, Section III describes the design and data acquisition process of the simulated fatigue driving experiments, Section IV establishes the fatigue detection model based on deep convolutional neural networks, Section V presents the detection results of the model, Section VI discusses the findings, and Section VII concludes the paper.

II. RELATED WORK

Over the past few decades, with the increasing demands for vehicle safety and intelligence, driver fatigue detection has become a research focus in the scientific community and various commercial companies. Existing fatigue detection methods [3], [23], [24] can be classified into three basic approaches based on different data sources: vehicle motion parameters, driver behavioral features, and driver physiological signals, as shown in Table I. Furthermore, fatigue detection methods based on the fusion of multiple information sources have also been proposed and have gained significant attention from researchers. This chapter reviews the development of these three basic approaches and summarizes the latest achievements in fatigue detection based on multi-source information fusion method.

When drivers enter a state of fatigue, their ability to control the vehicle decreases, and there are significant changes in the vehicle's motion compared to when they are alert. Many researchers have attempted to detect driver fatigue by collecting parameters such as vehicle speed [25], acceleration [26], and lane deviation [27]. However, with the emergence of advanced driver assistance systems and autonomous driving, drivers have less control over the vehicle, making it difficult for vehicle motion parameters to accurately reflect their true fatigue state. Therefore, considering the future development prospects of fatigue detection methods based on vehicle motion parameters, we did not use vehicle motion parameters for fatigue detection in this study.

Physiological signals, which can intuitively reflect the driver's fatigue state, have always been a focus of fatigue detection research. Fatigue-related physiological signals include EEG [28], heart rate [29], and ECG [30]. Among them, EEG is considered the gold standard for fatigue detection [31]. Researchers [32] have used a spatiotemporal convolutional neural network based on EEG signals for driver

TABLE I
DESCRIPTION OF DRIVER FATIGUE DETECTION SYSTEMS

Detection System	Input features	Interference factors	Accuracy	Practicality
Based on vehicle motion parameters	Speed, Lane position, Acceleration, etc.	Vehicle model, Driving conditions, etc.	General	Good
Based on driver behavioral features	PERCLOSE, Yawning frequency, Head pose, etc.	Illumination, Occlusion, etc.	Good	Good
Based on physiological signals	Heart rate, Electroencephalogram, Electroencephalogram, etc.	interference signals, human bodily motions, etc.	Excellent	Poor

fatigue detection, achieving an accuracy rate of 97.37%. Another study extracted 13 highly correlated features from the driver's ECG and trained three classifiers, namely support vector machine, K-nearest neighbors, and ensemble classifiers, achieving an overall fatigue recognition accuracy of 96.6% [33]. Despite the high accuracy of detecting driver fatigue using physiological signals such as EEG and ECG, the requisite equipment needs to be placed close to the driver's head or chest, which is very intrusive and may cause discomfort. Consequently, the practicality of using such physiological signals for fatigue detection is relatively low. With the rapid development of wearable devices, the collection of some common and basic physiological signals, such as heart rate and blood oxygen saturation, has become more comfortable and convenient. It provides a feasible solution for fatigue detection systems based on physiological signals with broad application prospects. Therefore, future research should focus on using these easily collected physiological signals to detect fatigue.

The extraction of driver behavior features through video analysis offers a non-contact approach, particularly with the rapid development of computer vision technology. This fatigue detection method has attracted widespread attention among researchers. Various fatigue feature indicators have been proposed, including blink frequency [34], head pose angles [12], and mouth opening angles [35]. Among them, PERCLOS (Percentage of Eye Closure) is considered one of the most effective behavior features [36]. To improve detection accuracy, researchers tend to combine multiple fatigue feature indicators to discriminate driver fatigue states. Some studies use a convolutional neural network to detect driver fatigue based on PERCLOS and mouth opening frequency [37]. Other studies combine five indicators, such as single eye closure time, blink frequency, and eye closure rate, to assess driver fatigue [38]. Current research on driver behavior fatigue features primarily focuses on the mouth and eyes [19], leaving the potential relationship between other facial regions and fatigue unclear. However, studies [39] have shown that these features, which are based on local information, exhibit significant inter-individual variability due to their limited information content. In contrast, complete facial information includes more potential fatigue-related features, such as facial expressions and associated muscle movements, as demonstrated in related

research [40]. Therefore, it is necessary to investigate the use of complete facial information for fatigue detection.

In order to fully exploit the redundancy and complementarity of information, researchers have combined the above three methods and proposed a fatigue detection method based on multi-source information fusion [24]. Commonly used information fusion methods include deep learning techniques [41], rough set theory [42], dynamic Bayesian networks [43], and D-S evidence theory [44]. In recent years, with the excellent performance of deep learning techniques in various fields, an increasing number of studies have adopted deep learning for information fusion [19]. Despite the powerful capabilities of deep learning in extracting features, existing research typically uses traditional hand-picked features as model inputs, using deep learning as a classifier rather than a feature extractor. For example, study extracted PERCLOS and heart rate, using a convolutional neural network for information fusion to detect driver fatigue states [22]. The work reported in [45] used a recurrent neural network to fuse three features: heart rate, eye openness level, and mouth openness level. In another work [46], 29 handcrafted fatigue features were extracted and a single-layer neural network was used for fatigue state detection. This reliance on handcrafted features significantly influences the fatigue detection performance. Handcrafted features are determined based on prior knowledge and, while they offer good interpretability, they struggle to capture deep-level fatigue features, affecting the accuracy and generalization performance of fatigue detection. In addition, the use of hand-crafted features introduces real-time performance issues, as extracting these features requires a long-time interval. For example, the calculation of PERCLOS values requires a 1 minute interval [47], while the calculation of heart rate variability requires between 1 and 5 minutes [23].

As described above, the method of fatigue detection for drivers through the fusion of multi-source information, including physiological signals and behavior features, aligns with the future development direction. Therefore, in this study, we have adopted such an approach to establish fatigue detection model. To overcome the challenges associated with collecting physiological signals, we have discarded signals that are difficult to collect, such as EEG and ECG, and instead focused on signals that can be collected using wearable devices, such

as heart rate, electrodermal activity and blood volume pulse. In the context of driver behavior features, to capture as many potential fatigue-related features as possible, we used whole face images as input to the detection model. We did not perform any manual feature extraction, but directly fed physiological signals and facial images into a deep convolutional neural network for automatic feature extraction. To validate the rationality of using the neural network as a feature extractor instead of traditional hand-picked features, we performed a visual analysis of the model. It also helps to reveal the potential relationships between facial regions and the expression of fatigue.

III. MATERIALS AND METHODS

A. Participants

The study [48] showed that young drivers are a high-risk group for fatigued driving. Therefore, this experiment primarily focused on young drivers, with a total of 21 participants (mean age $\pm$ SD: 23.04 ± 1.1 years). The ratio of male to female of the drivers was 2:1, consistent with the gender distribution of drivers in China. All participants had normal vision (including corrected vision) and normal hearing, and none reported experiencing dizziness from 3D animations. Prior to participating in the experiment, participants were prohibited from consuming alcohol, coffee, tea, or other stimulating beverages. The data of all participants underwent processing in accordance with the Helsinki Declaration.

B. Equipment

In this study, we conducted a simulated fatigue driving experiment for data collection. The driving simulator used in the experiment is a high-fidelity, immersive car simulator with three degrees of freedom. The simulator is equipped with features similar to those of a real car, including a steering wheel, brake, clutch, etc., and can simulate driving conditions such as vibration and inclination. The rear-view mirror of the car simulator, together with the surrounding array of six electronic screens, provides the driver with a 360° field of view, as shown in Fig. 2 (a).

Two types of data collection devices were used, as illustrated in Fig. 2(b). A high-resolution camera was installed in front of the driver to capture facial images throughout the driving session. The drivers are required to wear smart watch throughout the entire driving session to record three physiological signals: heart rate, electrodermal activity, and blood volume pulse. Prior research [18], [49] has demonstrated that these signals can effectively reflect the driver's fatigue state. The types of data collected and their respective sampling frequencies are summarized in Table II.

C. Protocol

We divided the formal experiment into two groups: the alertness group and the fatigue group, with each participant needed to complete both sets of experiments. The experiment with the alertness group began at 9:00 am, with participants required to sleep 9 hours a day, from 22:00 pm to 7:00 am. The fatigued group started at 3:00 pm, when according to

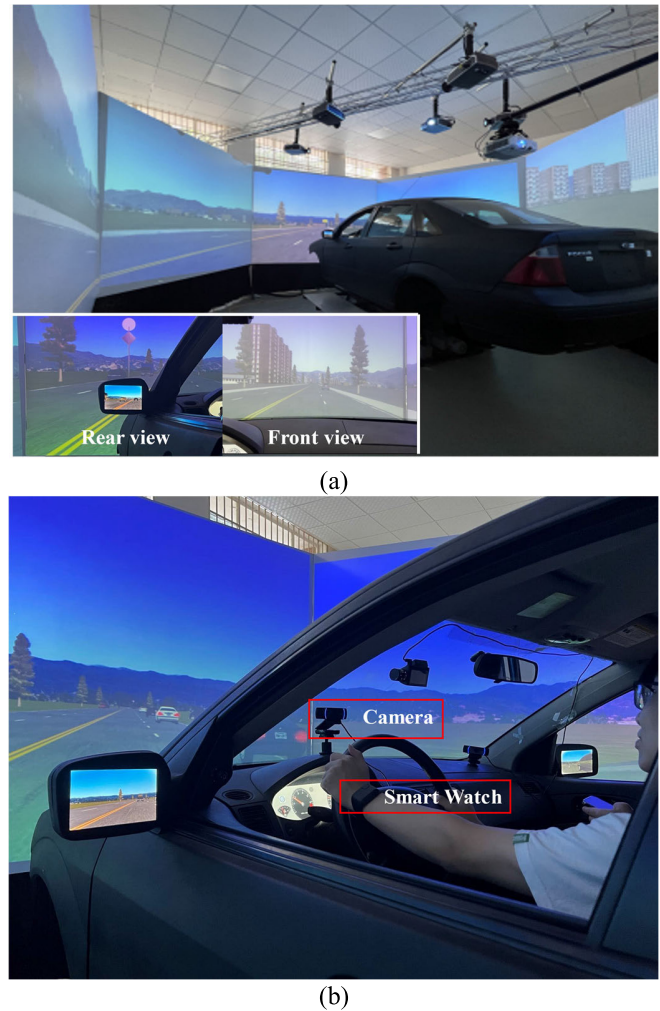


Fig. 2. (a) Appearance of simulated driving simulator and Driver's view. (b) Data collection devices.

TABLE II
DETAILS OF EXPERIMENTAL DATA COLLECTION

Device	Signal/Data	Sampling rate
Logitech C920	Video	30Hz
Empatica E4	Heart rate (HR)	1Hz
	Electrodermal activity (EDA)	4Hz
	Blood volume pulse (BVP)	32Hz

the circadian rhythm drivers are at a high risk of drowsiness [50]. We also use sleep deprivation to induce fatigue, with participants instructed to get only 5 hours of sleep before the fatigue group experiment, from 2:00 am to 7:00 am [45], [51]. The experimental set-up allowed participants to complete only one group's experiment on each experimental day.

Prior to each simulated driving session, all participants were required to operate the driving simulator and drive a certain distance until they became familiar with the virtual road scenarios and the operation of the driving simulator. During this period, participants were allowed to adjust the seat height and backrest angle to ensure driving comfort and maximize the resemblance to real-life driving experience.

In both groups of experiments, participants were required to complete a driving route with a total length of 80 km.

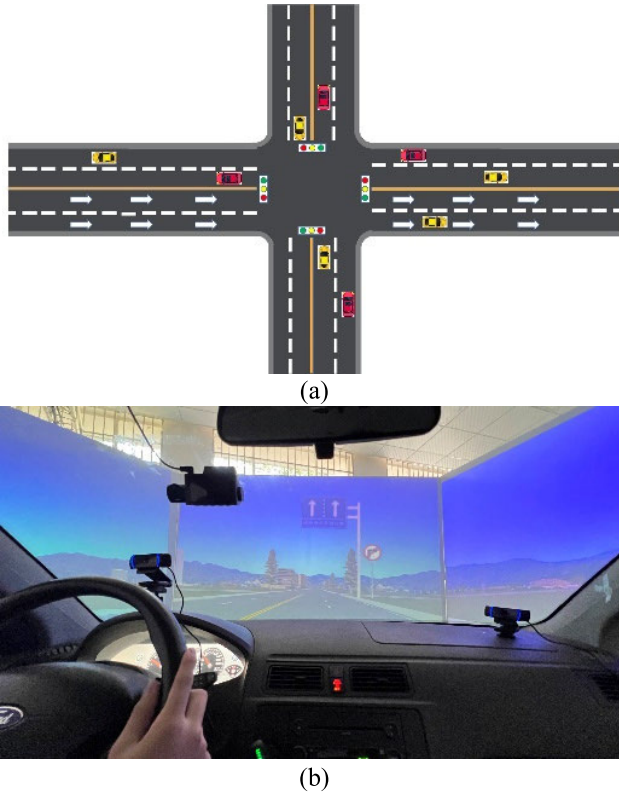


Fig. 3. (a) Illustration of partial driving routes. (b) Simulated experimental scenario.

The driving route consisted of a four-lane, two-way road, primarily composed of straight sections, along with 8 intersections and 2 curves, as depicted in Fig. 3 (a). In order to simulate complex real-world driving scenarios, the simulated driving environment includes the presence of 2 or 3 additional vehicles, serving as other traffic participants. It also encompasses road markings, traffic signs, traffic lights, and other traffic infrastructure as shown in Fig. 3 (b). During the experiment, participants were required to follow traffic signals and regulations indicated by traffic signs, while also avoiding potential collisions with other vehicles. The maximum speed limit was set at 80 kilometers per hour, which is the maximum speed allowed on urban roads in China [52]. Consequently, the driving time in the simulator was at least one hour for each group of experiment.

At the end of each simulated driving experiment, participants are required to subjectively evaluate their fatigue states based on the recorded videos from the camera. They need to label specific time segments as “fatigue” or “alert” states. If a participant is unable to accurately determine their own state during certain time segments, those data will be excluded. Additionally, to avoid potential fluctuations in driver’s state at the beginning or near the end of the experiment, data collected during the first and last minute after the start and before the end of the experiment will not be considered, in order to enhance the reliability of the data.

IV. DATA ANALYSIS AND MODELING

A. Data Preprocess

To achieve better recognition results, we performed pre-processing on the collected videos and physiological signals.

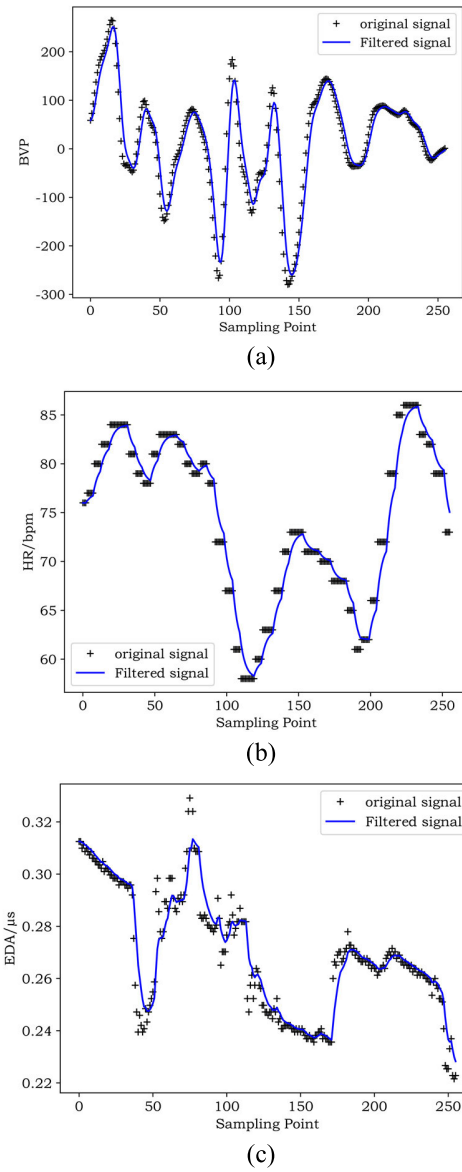


Fig. 4. Comparison of different physiological signals before and after filtering: (a) blood volume pulse, (b) heart rate, (c) electrodermal activity.

Denosing is crucial for processing raw physiological signals as they often contain various types of noise introduced by the data acquisition devices and human body activities. To meet the practical requirements of the fatigue detection system, Kalman filtering [53] is used for signal processing in this study. The advantage of Kalman filtering is that it only needs to store the state estimate from the previous moment, resulting in low memory consumption, simple computation and fast speed, making it suitable for real-time detection. The filtered physiological signals are smoother compared to the noisy raw signals, without frequent fluctuations or abrupt changes, as shown in Fig. 4.

Since facial features of the driver are primarily used for fatigue detection through video analysis, we employed the Multi-Task Cascaded Convolutional Networks (MTCNN) for face tracking and detection. This algorithm [54] utilizes a cascade of classifiers with candidate bounding boxes, allowing for fast and accurate face detection using three cascaded

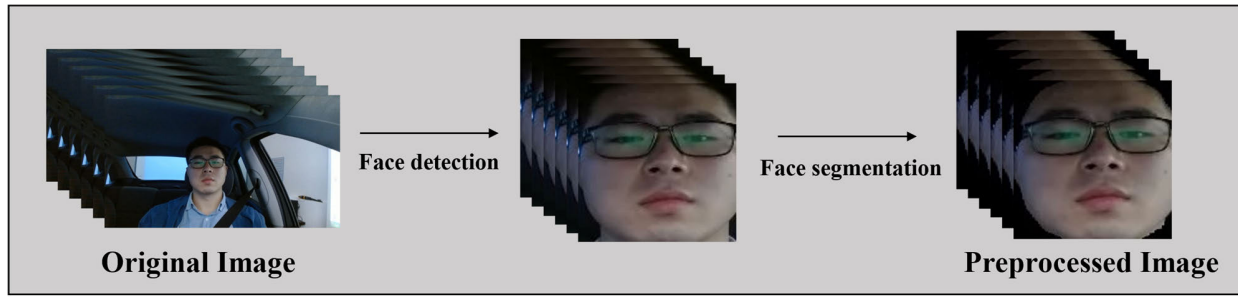


Fig. 5. Image preprocessing workflow.

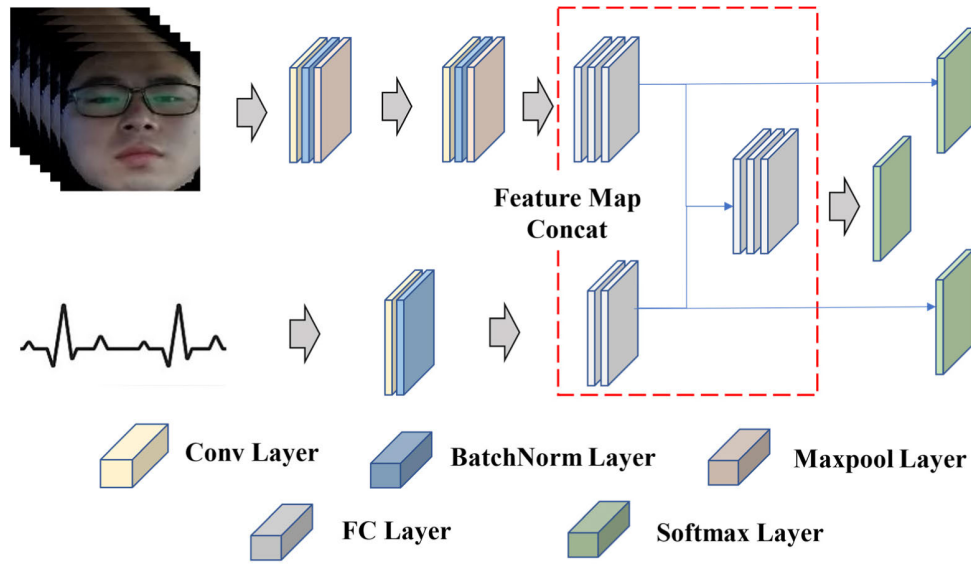


Fig. 6. Architecture of the fatigue detection network model.

networks. After obtaining the face bounding box, we applied Linknet [55], an efficient semantic segmentation algorithm, to perform semantic segmentation on the entire face. Through these steps, we retained only the driver's face in the video, removing unnecessary background information, allowing the model to focus on detecting facial features. The specific processing steps are illustrated in Fig. 5. Additionally, to reduce computational load, we resized the images to 64×64 dimensions.

B. Model

Deep learning techniques, particularly convolutional neural networks (CNNs), have emerged as the future direction in driver fatigue detection, especially for analyzing video sequences [19]. Research has shown that CNNs outperform recurrent neural networks and support vector machines in classification tasks [41], [56]. Therefore, we developed a fatigue detection model that incorporates multi-source information fusion using a deep CNN for physiological signals and videos. Additionally, to understand the contributions of different signal sources to the fatigue detection results, we built separate fatigue detection models based on physiological signals and videos. The specific model frameworks are illustrated in Fig. 6.

The upper part of Fig. 6 illustrates the fatigue detection model based solely on video sequences. We utilize a

three-dimensional convolutional neural network (3D CNN) to extract temporal information from the video sequences. The model consists of 2 3D convolutional layers, 2 normalization layers, 2 max pooling layers, and 3 fully connected layers. The kernel size for the first convolutional layer is $5 \times 5 \times 5$, while the second convolutional layer has a kernel size of 3×3 .

The lower part of Fig. 6 shows the fatigue detection model based on physiological signals. Since physiological signals are purely temporal, we employ a one-dimensional convolutional neural network (1D CNN) for feature extraction. The model consists of 1 convolutional layer, 1 normalization layer, and 2 fully connected layers. The kernel size for the convolutional layer is 3.

The middle part of the diagram presents the fatigue detection model that combines video sequences and physiological signals. After obtaining the feature maps from the last fully connected layer of the individual detection models, we concatenate them and input the concatenated features into a 3-layer fully connected layer for information fusion.

Additionally, we incorporate Dropout layers after each fully connected layer to prevent overfitting [57]. The dropout probability is set to 0.4. The Rectified Linear Unit (ReLU) function serves as the activation function for the convolutional and fully connected layers. All model outputs are normalized through

TABLE III
MODEL DETECTION ACCURACY WITH TEN-FOLD CROSS-VALIDATION

Input	Indicator	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Fold 6	Fold 7	Fold 8	Fold 9	Fold 10	Mean
Video sequence Only	Accuracy	89.44%	88.06%	88.89%	89.17%	88.61%	87.78%	90.28%	89.72%	89.44%	90.56%	89.20%
	Specificity	92.54%	88.27%	92.04%	87.65%	89.90%	89.45%	91.50%	92.39%	92.04%	93.47%	90.93%
	Sensitivity	85.53%	87.8%	84.91%	90.40%	87.04%	85.44%	82.5%	86.88%	86.16%	86.96%	86.36%
Physiological signals Only	Accuracy	62.78%	58.33%	61.67%	59.72%	60.28%	59.44%	62.78%	62.22%	61.11%	59.17%	60.75%
	Specificity	89.30%	83.57%	88.37%	84.79%	86.73%	85.05%	91.20%	91.04%	87.91%	83.89%	87.19%
	Sensitivity	23.45%	21.77%	22.07%	21.68	22.82%	21.92%	20.14%	20.94%	21.38%	21.48%	21.77%
Multi-source information	Accuracy	91.07%	94.05%	93.45%	92.86%	91.67%	92.26%	95.24%	94.64%	93.06%	93.18%	93.15%
	Specificity	91.09%	94.95%	95.00%	93.00%	91.18%	93.46%	98.11%	96.12%	93.02%	94.44%	94.04%
	Sensitivity	91.04%	92.75%	91.17%	92.65%	92.41%	90.16%	90.32%	92.31%	93.10%	91.18%	91.71%

a Softmax layer. As driver fatigue detection is essentially a classification task, the cross-entropy loss function has shown good performance for such tasks. Therefore, we employ the cross-entropy loss function for model training, with its specific computational expression depicted in equation (1).

$$J(\theta) = -\frac{1}{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

In equation (1), N denotes the total number of samples, θ represents the parameters of the model, $y_i \in \{0, 1\}$ signifies the true label of the i sample, and $\hat{y}_i \in [0, 1]$ denotes the predicted label of the i sample by the model.

V. RESULT

A. Result of Driver Fatigue Detection

In order to maximize the real-time performance of the fatigue detection system and considering the data volume collected per second, we set the computational time window to 3 seconds, which is the shortest window known to us in the field of driver state detection. Both the video sequences and physiological signals are inputted in segments of 3 seconds. Finally, we got a total of 45,480 data segments, including 19,296 fatigued segments and 26,184 non-fatigued segments. After preprocessing, the video sequence has an image size of 64×64 , resulting in input data size of $90 \times 64 \times 64$. For the combination of EDA, HR, and BVP signals, the input data size for physiological signals is 3×192 .

We divided the dataset into 90% for training and 10% for testing. The Adam optimizer was used for training with a batch size of 16 and a learning rate of 0.001. A total of 150 epochs were trained. All model training was conducted on a Windows system, with a 13th generation i9 CPU and an RTX4090 GPU. To reduce the impact of imbalanced dataset distribution on detection accuracy, we employed ten-fold cross-validation for training and testing the three models separately. In addition, to comprehensively reflect the overall detection performance of the model, we used three evaluation metrics: accuracy, specificity and sensitivity. The evaluation metrics are defined as shown in equations (2)–(4), where TP represents the number of true positive samples, TN represents the number of true negative samples, FP represents the number of false

positive samples, and FN represents the number of false negative samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (3)$$

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

The overall performance evaluation of the three methods is presented in Table III. The fatigue detection model based on multi-source information fusion achieved the best performance across all three metrics, with an average detection accuracy of 93.15%. The model also showed a balanced ability to discriminate between fatigued and non-fatigued states (specificity = 94.04%, sensitivity = 91.71%). Among the models based on single sources of information, the video-based fatigue detection model performed better, with an average detection accuracy of 89.20% (specificity = 90.93%, sensitivity = 86.36%). In contrast, the fatigue detection model based on physiological signals had a lower detection accuracy, with an average accuracy of 60.75%, and was prone to missed detections (specificity = 87.19%, sensitivity = 21.77%).

B. Visual Analysis of Fatigue Detection Model

According to Table III, in the fatigue detection model based on multi-source information, the video sequence plays a crucial role in determining the detection results. However, the black box nature of deep learning makes it difficult to intuitively understand the specific fatigue features extracted by the model from facial images. Therefore, we employed the layercam [58] method for visual analysis of the model. This method can generate class activation maps for any convolutional layer in the model without the need for retraining, revealing the importance weights of different parts of the image on the detection results.

Using the trained model, we conducted detection experiments on different driver fatigue states, head poses, and glasses occlusion scenarios. The class activation maps for the last convolutional layer of the model are shown in Fig. 7. We observed that the model focuses on the eyebrows, eyes, mouth, and

TABLE IV
DETECTION PERFORMANCE COMPARISON OF RELATED WORK

Reference	Granularity extraction	Input features	Time window	Accuracy
[21]	Eyes, Head, Lane	Gaze direction, Head pose, Lane positions, Lane departures, Lane changes	300s	92%
[22]	Eyes, Physiological signal	HR, PERCLOS	60s	86.4%
[60]	Eyes	Blink rate	600s	65.5%
[61]	Eyes, Mouth, Nose	EAR, MAR, MVE, Centre coordinate of pupil	5s	79.9%
[62]	Eyes	Eyes move	30s	90.9%
Ours	Entire face, Physiological signal	Neural network extraction of features	3s	93.15%



Fig. 7. Visualization of class activation maps. (a) and (b) source images of different driver fatigue states. (b) and (e) grayscale class activation maps. (c) and (f) colored class activation maps on source image.

muscles around the nose area of the face, indicating their strong correlation with driver fatigue states. Among these, the eyes receive the most attention from the model, and the state of the eyes in each video sequence has the greatest impact on the detection results.

VI. DISCUSSION

The fatigue detection model based on a single physiological signal information source exhibits a lower detection accuracy, merely at 60.75%, when compared to the detection model based on video sequences. This indicates that although using physiological signals for fatigue detection is considered an accurate approach, it is challenging for physiological signals

to real-time reflect the true fatigue state of drivers within short computational time window. Additionally, this discrepancy may be attributed to individual differences in drivers' physiological signals, as research [59] has shown that the unique nature of fatigue driving (where drivers may be fatigued but strive to maintain alertness to respond to potential traffic risks) could magnify these differences. On the other hand, the method based on video sequences for fatigue detection demonstrates unique advantages in real-time fatigue detection. This may be because drivers exhibit characteristic behaviors such as eye closure and yawning during fatigue, which are easily captured within short time frames.

Interestingly, despite the differences in the performance of fatigue detection methods based on single information sources, the model that utilized multi-source information fusion still achieved higher detection accuracy. This suggests that deep learning information integration can effectively exploit the correlation and complementarity of different fatigue features, thereby improving the accuracy of fatigue detection. Therefore, future research should consider incorporating as many fatigue-related features as possible.

Compared to other related works listed in Table IV, our detection method achieves the best performance in a shorter time. This result shows that our method of automatically extracting features using neural networks has significant advantages over traditional those hand-picked features. End-to-end fatigue detection by deep learning models has shown immense potential. To realize this potential, there is an urgent need to establish a more diverse and comprehensive dataset that encompasses a wider range of individuals and is more representative of real-world driving scenarios. This would help to address the issue of model generalization that arises in the practical application of fatigue detection.

The visualization analysis of our model shows that the detection model focuses primarily on the eye and mouth regions. This finding is consistent with existing research and validates the rationality and reliability of the model. Interestingly, the model also highlights fatigue features in the eyebrow and nose tip regions. This may be due to general facial muscle movements, such as blinking and yawning, which can affect these areas. As handcrafted features are typically unable to

capture these complex variations, these regions have not been considered in existing studies. It demonstrates the superiority of our method in extracting potential fatigue features.

By the powerful feature extraction capabilities of deep learning models, we have revealed potential relationships between facial regions and fatigue. The results show that the eyes, eyebrows, mouth and nose are all closely associated with fatigue, with the eyes being the most strongly correlated. However, other regions of the face have a relatively weaker relationship with fatigue. Future research could focus on more precise segmentation of these regions, which would allow more effective use of facial fatigue features.

VII. CONCLUSION

This paper proposes a fatigue detection method based on the fusion of physiological signals and facial image information, which does not rely on traditional hand-picked features but uses neural networks for automatic feature extraction. It achieves a detection accuracy of 93.15% (specificity = 94.04%, sensitivity = 91.71%) within a 3-second time window, outperforming existing methods in terms of both accuracy and real-time performance. To address the interpretability issues associated with deep learning, we performed a visualization analysis of the model. The results show that our model extracts fatigue features from key regions such as the eyes and mouth, as well as capturing potential fatigue features from a wider range of regions, demonstrating the rationality and effectiveness of our method. As another major contribution of this paper, the results of the visualization analysis also reveal the potential relationship between facial regions and fatigue expression. In particular, we found that the eyebrow and nose regions are closely related to fatigue, which contributes to more accurate segmentation of facial regions. Future work includes improvements to deep learning models to improve detection performance. In addition, a larger dataset is needed to address the generalization issues of deep learning-based fatigue detection approach.

REFERENCES

- [1] C. Zhao, C. Zheng, M. Zhao, Y. Tu, and J. Liu, "Multivariate autoregressive models and kernel learning algorithms for classifying driving mental fatigue based on electroencephalographic," *Exp. Syst. Appl.*, vol. 38, no. 3, pp. 1859–1865, Mar. 2011, doi: [10.1016/j.eswa.2010.07.115](#).
- [2] T. Akerstedt, "Consensus statement: Fatigue and accidents in transport operations," *J. Sleep Res.*, vol. 9, no. 4, p. 395, Dec. 2000, doi: [10.1046/j.1365-2869.2000.00228.x](#).
- [3] Y. Dong, Z. Hu, K. Uchimura, and N. Murayama, "Driver inattention monitoring system for intelligent vehicles: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 2, pp. 596–614, Jun. 2011, doi: [10.1109/TITS.2010.2092770](#).
- [4] K. Zeeb, A. Buchner, and M. Schrauf, "What determines the take-over time? An integrated model approach of driver take-over after automated driving," *Accident Anal. Prevention*, vol. 78, pp. 212–221, May 2015, doi: [10.1016/j.aap.2015.02.023](#).
- [5] O. Jarosch, S. Paradies, D. Feiner, and K. Bengler, "Effects of non-driving related tasks in prolonged conditional automated driving—A wizard of Oz on-road approach in real traffic environment," *Transp. Res. F, Traffic Psychol. Behaviour*, vol. 65, pp. 292–305, Aug. 2019, doi: [10.1016/j.trf.2019.07.023](#).
- [6] T. Vogelpohl, M. Kühn, T. Hummel, and M. Vollrath, "Asleep at the automated wheel—Sleepiness and fatigue during highly automated driving," *Accident Anal. Prevention*, vol. 126, pp. 70–84, May 2019, doi: [10.1016/j.aap.2018.03.013](#).
- [7] T. Tuncer, S. Dogan, F. Ertam, and A. Subasi, "A dynamic center and multi threshold point based stable feature extraction network for driver fatigue detection utilizing EEG signals," *Cognit. Neurodynamics*, vol. 15, no. 2, pp. 223–237, Apr. 2021, doi: [10.1007/s11571-020-09601-w](#).
- [8] G. Du, S. Long, C. Li, Z. Wang, and P. X. Liu, "A product fuzzy convolutional network for detecting driving fatigue," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4175–4188, Jun. 2023, doi: [10.1109/TCYB.2021.3123842](#).
- [9] F. Zhang, J. Su, L. Geng, and Z. Xiao, "Driver fatigue detection based on eye state recognition," in *Proc. Int. Conf. Mach. Vis. Inf. Technol. (CMVIT)*, Singapore, Feb. 2017, pp. 105–110, doi: [10.1109/CMVIT.2017.25](#).
- [10] Y. Qiao, K. Zeng, L. Xu, and X. Yin, "A smartphone-based driver fatigue detection using fusion of multiple real-time facial features," in *Proc. 13th IEEE Annu. Commun. Netw. Conf. (CCNC)*, Las Vegas, NV, USA, Jan. 2016, pp. 230–235, doi: [10.1109/CCNC.2016.7444761](#).
- [11] A. Dasgupta, A. George, S. L. Happy, and A. Routray, "A vision-based system for monitoring the loss of attention in automotive drivers," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 4, pp. 1825–1838, Dec. 2013.
- [12] R. O. Mbouna, S. G. Kong, and M.-G. Chun, "Visual analysis of eye state and head pose for driver alertness monitoring," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 3, pp. 1462–1469, Sep. 2013, doi: [10.1109/TITS.2013.2262098](#).
- [13] B. Cyganek and S. Gruszczyński, "Hybrid computer vision system for drivers' eye recognition and fatigue monitoring," *Neurocomputing*, vol. 126, pp. 78–94, Feb. 2014, doi: [10.1016/j.neucom.2013.01.048](#).
- [14] G. Sikander and S. Anwar, "Driver fatigue detection systems: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 6, pp. 2339–2352, Jun. 2019, doi: [10.1109/TITS.2018.2868499](#).
- [15] B. Mandal, L. Li, G. S. Wang, and J. Lin, "Towards detection of bus driver fatigue based on robust visual analysis of eye state," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 3, pp. 545–557, Mar. 2017, doi: [10.1109/tits.2016.2582900](#).
- [16] Y. Ji, S. Wang, Y. Zhao, J. Wei, and Y. Lu, "Fatigue state detection based on multi-index fusion and state recognition network," *IEEE Access*, vol. 7, pp. 64136–64147, 2019, doi: [10.1109/ACCESS.2019.2917382](#).
- [17] M. Shahverdy, M. Fathy, R. Berangi, and M. Sabokrou, "Driver behavior detection and classification using deep convolutional neural networks," *Exp. Syst. Appl.*, vol. 149, Jul. 2020, Art. no. 113240.
- [18] S.-Y. Han, N.-S. Kwak, T. Oh, and S.-W. Lee, "Classification of pilots' mental states using a multimodal deep learning network," *Biocybernetics Biomed. Eng.*, vol. 40, no. 1, pp. 324–336, Jan. 2020, doi: [10.1016/j.bbe.2019.12.002](#).
- [19] F. Liu, D. Chen, J. Zhou, and F. Xu, "A review of driver fatigue detection and its advances on the use of RGB-D camera and deep learning," *Eng. Appl. Artif. Intell.*, vol. 116, Nov. 2022, Art. no. 105399, doi: [10.1016/j.engappai.2022.105399](#).
- [20] K. Dwivedi, K. Biswaranjan, and A. Sethi, "Drowsy driver detection using representation learning," in *Proc. IEEE Int. Advance Comput. Conf. (IACC)*, Feb. 2014, pp. 995–999, doi: [10.1109/IADCC.2014.6779459](#).
- [21] B. Bakker et al., "A multi-stage, multi-feature machine learning approach to detect driver sleepiness in naturalistic road driving conditions," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4791–4800, May 2022, doi: [10.1109/TITS.2021.3090272](#).
- [22] G. Du, L. Zhang, K. Su, X. Wang, S. Teng, and P. X. Liu, "A multimodal fusion fatigue driving detection method based on heart rate and PERCLOS," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21810–21820, Nov. 2022, doi: [10.1109/TITS.2022.3176973](#).
- [23] K. Lu, A. S. Dahlman, J. Karlsson, and S. Candefjord, "Detecting driver fatigue using heart rate variability: A systematic review," *Accident Anal. Prevention*, vol. 178, Dec. 2022, Art. no. 106830, doi: [10.1016/j.aap.2022.106830](#).
- [24] A. Nemcová et al., "Multimodal features for detection of driver stress and fatigue: Review," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 6, pp. 3214–3233, Jun. 2021, doi: [10.1109/TITS.2020.2977762](#).
- [25] W. Qi, Z. Wang, R. Tang, and L. Wang, "Driving risk detection model of deceleration zone in expressway based on generalized regression neural network," *J. Adv. Transp.*, vol. 2018, pp. 1–8, Oct. 2018, doi: [10.1155/2018/8014385](#).
- [26] J. Xi, S. Wang, T. Ding, J. Tian, H. Shao, and X. Miao, "Detection model on fatigue driving behaviors based on the operating parameters of freight vehicles," *Appl. Sci.*, vol. 11, no. 15, p. 7132, Aug. 2021, doi: [10.3390/app11157132](#).

- [27] A. Assuncao, A. Aquino, R. Câmara de M. Santos, R. Guimaraes, and R. Oliveira, "Vehicle driver monitoring through the statistical process control," *Sensors*, vol. 19, no. 14, p. 3059, Jul. 2019, doi: [10.3390/s19143059](https://doi.org/10.3390/s19143059).
- [28] C. Fan, Y. Peng, S. Peng, H. Zhang, Y. Wu, and S. Kwong, "Detection of train driver fatigue and distraction based on forehead EEG: A time-series ensemble learning method," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 13559–13569, Aug. 2022, doi: [10.1109/TITS.2021.3125737](https://doi.org/10.1109/TITS.2021.3125737).
- [29] M. Patel, S. K. L. Lal, D. Kavanagh, and P. Rossiter, "Applying neural network analysis on heart rate variability data to assess driver fatigue," *Exp. Syst. Appl.*, vol. 38, no. 6, pp. 7235–7242, Jun. 2011, doi: [10.1016/j.eswa.2010.12.028](https://doi.org/10.1016/j.eswa.2010.12.028).
- [30] R. Fu and H. Wang, "Detection of driving fatigue by using noncontact EMG and ECG signals measurement system," *Int. J. Neural Syst.*, vol. 24, no. 3, May 2014, Art. no. 1450006, doi: [10.1142/s0129065714500063](https://doi.org/10.1142/s0129065714500063).
- [31] I. Stancin, M. Cifrek, and A. Jovic, "A review of EEG signal features and their application in driver drowsiness detection systems," *Sensors*, vol. 21, no. 11, p. 3786, May 2021, doi: [10.3390/s21113786](https://doi.org/10.3390/s21113786).
- [32] Z. Gao et al., "EEG-based spatio-temporal convolutional neural network for driver fatigue evaluation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2755–2763, Sep. 2019, doi: [10.1109/TNNLS.2018.2886414](https://doi.org/10.1109/TNNLS.2018.2886414).
- [33] S. Murugan, J. Selvaraj, and A. Sahayadhas, "Detection and analysis: Driver state with electrocardiogram (ECG)," *Phys. Eng. Sci. Med.*, vol. 43, no. 2, pp. 525–537, Jun. 2020, doi: [10.1007/s13246-020-00853-8](https://doi.org/10.1007/s13246-020-00853-8).
- [34] J. He, W. Choi, Y. Yang, J. Lu, X. Wu, and K. Peng, "Detection of driver drowsiness using wearable devices: A feasibility study of the proximity sensor," *Appl. Ergonom.*, vol. 65, pp. 473–480, Nov. 2017, doi: [10.1016/j.apergo.2017.02.016](https://doi.org/10.1016/j.apergo.2017.02.016).
- [35] L. Mou et al., "Isotropic self-supervised learning for driver drowsiness detection with attention-based multimodal fusion," *IEEE Trans. Multimedia*, vol. 25, pp. 529–542, 2023, doi: [10.1109/TMM.2021.3128738](https://doi.org/10.1109/TMM.2021.3128738).
- [36] Q. Zhuang, Z. Kehua, J. Wang, and Q. Chen, "Driver fatigue detection method based on eye states with pupil and iris segmentation," *IEEE Access*, vol. 8, pp. 173440–173449, 2020, doi: [10.1109/ACCESS.2020.3025818](https://doi.org/10.1109/ACCESS.2020.3025818).
- [37] H. Li, Y. Wang, and Y. Nan, "Motion fatigue state detection based on neural networks," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–10, Mar. 2022, doi: [10.1155/2022/9602631](https://doi.org/10.1155/2022/9602631).
- [38] H. Jia, Z. Xiao, and P. Ji, "Real-time fatigue driving detection system based on multi-module fusion," *Comput. Graph.*, vol. 108, pp. 22–33, Nov. 2022, doi: [10.1016/j.cag.2022.09.001](https://doi.org/10.1016/j.cag.2022.09.001).
- [39] M. Ingre, T. Åkerstedt, B. Peters, A. Anund, and G. Kecklund, "Subjective sleepiness, simulated driving performance and blink duration: Examining individual differences," *J. Sleep Res.*, vol. 15, no. 1, pp. 47–53, Mar. 2006, doi: [10.1111/j.1365-2869.2006.00504.x](https://doi.org/10.1111/j.1365-2869.2006.00504.x).
- [40] G. Du, Z. Wang, B. Gao, S. Mumtaz, K. M. Abualnaja, and C. Du, "A convolution bidirectional long short-term memory neural network for driver emotion recognition," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4570–4578, Jul. 2021, doi: [10.1109/TITS.2020.3007357](https://doi.org/10.1109/TITS.2020.3007357).
- [41] S. Ansari, F. Naghdy, H. Du, and Y. N. Pahnwar, "Driver mental fatigue detection based on head posture using new modified reLU-BiLSTM deep neural network," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 10957–10969, Aug. 2022, doi: [10.1109/TITS.2021.3098309](https://doi.org/10.1109/TITS.2021.3098309).
- [42] Y. Du, Q. Hu, D. Chen, and P. Ma, "Kernelized fuzzy rough sets based yawn detection for driver fatigue monitoring," *Fundamenta Informaticae*, vol. 111, no. 1, pp. 65–79, Jan. 2011, doi: [10.3233/fi-2011-554](https://doi.org/10.3233/fi-2011-554).
- [43] G. Yang, Y. Lin, and P. Bhattacharya, "A driver fatigue recognition model based on information fusion and dynamic Bayesian network," *Inf. Sci.*, vol. 180, no. 10, pp. 1942–1954, May 2010, doi: [10.1016/j.ins.2010.01.011](https://doi.org/10.1016/j.ins.2010.01.011).
- [44] W. Zhu, H. Yang, Y. Jin, and B. Liu, "A method for recognizing fatigue driving based on Dempster-Shafer theory and fuzzy neural network," *Math. Problems Eng.*, vol. 2017, pp. 1–10, Jan. 2017, doi: [10.1155/2017/6191035](https://doi.org/10.1155/2017/6191035).
- [45] G. Du, T. Li, C. Li, P. X. Liu, and D. Li, "Vision-based fatigue driving recognition method integrating heart rate and facial features," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 5, pp. 3089–3100, May 2021, doi: [10.1109/TITS.2020.2979527](https://doi.org/10.1109/TITS.2020.2979527).
- [46] C. J. de Naurois, C. Bourdin, A. Stratulat, E. Diaz, and J.-L. Vercher, "Detection and prediction of driver drowsiness using artificial neural network models," *Accid. Anal. Prev.*, vol. 126, pp. 95–104, May 2019, doi: [10.1016/j.aap.2017.11.038](https://doi.org/10.1016/j.aap.2017.11.038).
- [47] D. F. Dinges and R. Grace, "PERCLOS: A valid psychophysiological measure of alertness as assessed by psychomotor vigilance," U.S. Dept. Transp., Federal Highway Admin., Washington, DC, USA, Publication Number FHWA-MCRT-98-006, 1998.
- [48] M. Chipman and Y. L. Jin, "Drowsy drivers: The effect of light and circadian rhythm on crash occurrence," *Saf. Sci.*, vol. 47, no. 10, pp. 1364–1370, Dec. 2009, doi: [10.1016/j.ssci.2009.03.005](https://doi.org/10.1016/j.ssci.2009.03.005).
- [49] Y. Zhao, K. Xie, Z. Zou, and J.-B. He, "Intelligent recognition of fatigue and sleepiness based on InceptionV3-LSTM via multi-feature fusion," *IEEE Access*, vol. 8, pp. 144205–144217, 2020, doi: [10.1109/ACCESS.2020.3014508](https://doi.org/10.1109/ACCESS.2020.3014508).
- [50] J. Horne and L. Reyner, "Vehicle accidents related to sleep: A review," *Occupational Environ. Med.*, vol. 56, no. 5, pp. 289–294, May 1999, doi: [10.1136/oem.56.5.289](https://doi.org/10.1136/oem.56.5.289).
- [51] N. Petrovsky et al., "Sleep deprivation disrupts prepulse inhibition and induces psychosis-like symptoms in healthy humans," *J. Neurosci.*, vol. 34, no. 27, pp. 9134–9140, Jul. 2014, doi: [10.1523/jneurosci.0904-14.2014](https://doi.org/10.1523/jneurosci.0904-14.2014).
- [52] (2004). *Road Traffic Safety Law of the People's Republic of China*. Accessed: Jul. 20, 2023. [Online]. Available: <https://flk.npc.gov.cn/detail2.html?ZmY4MDgwODE2ZjNjYmIzYzAxNmY0MGY1NjZjMTBIMjM>
- [53] E. Giraldo, A. J. den Dekker, and G. Castellanos-Dominguez, "Estimation of dynamic neural activity using a Kalman filter approach based on physiological models," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol.*, Aug. 2010, pp. 2914–2917, doi: [10.1109/IEMBS.2010.5626281](https://doi.org/10.1109/IEMBS.2010.5626281).
- [54] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, Oct. 2016, doi: [10.1109/LSP.2016.2603342](https://doi.org/10.1109/LSP.2016.2603342).
- [55] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4, doi: [10.1109/VCIP.2017.8305148](https://doi.org/10.1109/VCIP.2017.8305148).
- [56] H. A. Kassem, M. Chowdhury, and J. H. Abawajy, "Drivers fatigue level prediction using facial, and head behavior information," *IEEE Access*, vol. 9, pp. 121686–121697, 2021, doi: [10.1109/ACCESS.2021.3108561](https://doi.org/10.1109/ACCESS.2021.3108561).
- [57] C. Garbin, X. Zhu, and O. Marques, "Dropout vs. batch normalization: An empirical study of their impact to deep learning," *Multimedia Tools Appl.*, vol. 79, nos. 19–20, pp. 12777–12815, May 2020, doi: [10.1007/s11042-019-08453-9](https://doi.org/10.1007/s11042-019-08453-9).
- [58] P.-T. Jiang, C.-B. Zhang, Q. Hou, M.-M. Cheng, and Y. Wei, "LayerCAM: Exploring hierarchical class activation maps for localization," *IEEE Trans. Image Process.*, vol. 30, pp. 5875–5888, 2021, doi: [10.1109/TIP.2021.3089943](https://doi.org/10.1109/TIP.2021.3089943).
- [59] S. K. L. Lal and A. Craig, "A critical review of the psychophysiology of driver fatigue," *Biol. Psychol.*, vol. 55, no. 3, pp. 173–194, Feb. 2001, doi: [10.1016/S0301-0511\(00\)00085-5](https://doi.org/10.1016/S0301-0511(00)00085-5).
- [60] R. Ghoddoosian, M. Galib, and V. Athitsos, "A realistic dataset and baseline temporal model for early drowsiness detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Long Beach, CA, USA, Jun. 2019, pp. 178–187, doi: [10.1109/CVPRW.2019.00027](https://doi.org/10.1109/CVPRW.2019.00027).
- [61] N. N. Pandey and N. B. Muppalaneni, "Temporal and spatial feature based approaches in drowsiness detection using deep learning technique," *J. Real-Time Image Process.*, vol. 18, no. 6, pp. 2287–2299, Dec. 2021, doi: [10.1007/s11554-021-01114-x](https://doi.org/10.1007/s11554-021-01114-x).
- [62] Y. Xing, C. Lv, H. Wang, D. Cao, E. Velenis, and F.-Y. Wang, "Driver activity recognition for intelligent vehicles: A deep learning approach," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 5379–5390, Jun. 2019, doi: [10.1109/TVT.2019.2908425](https://doi.org/10.1109/TVT.2019.2908425).



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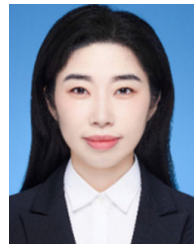
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